Siddhant Mahapatra

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Summary

Final-year Information Technology undergraduate focused on backend engineering and applied machine learning. Build REST APIs and full-stack applications in Python and JavaScript, with 7+ pull requests merged into JupyterLab and OpenSRE through maintainer code review. Comfortable with test automation, CI pipelines, and making decisions from data.

Experience

Open Source Contributor **Jan 2026 – May 2026**
JupyterLab & OpenSRE (Tracer) *Remote*

- Shipped a file-browser loading-animation feature to **JupyterLab** (**PR #18855**) in TypeScript; reviewed and merged by a core maintainer into the upstream codebase.
- Debugged and fixed failing Windows shell test commands in **OpenSRE's** CI pipeline (**PR #1755**), restoring cross-platform test reliability for their AI-driven SRE-agent toolkit.
- **7+ pull requests merged** across production open-source repos, working through code reviews on UI fixes, CI/test infrastructure, and dev tooling. View All Merged PRs

Data Science Trainee (Summer Training Program) **Jun 2025 – Aug 2025**
Bharati Vidyapeeth's College of Engineering *New Delhi*

- **Developed Three-D** | GitHub, a monocular depth estimation system that generates 3D representations from a single 2D image – no stereo or LiDAR required.
- Used **MiDaS** for the inference pipeline to extract structural depth, producing per-pixel disparity maps visualized as colored point clouds.

Projects

INE Price Tracker | *React, Node.js, Express, Playwright, Supabase (PostgreSQL)* | GitHub | Live **Sep 2026**

- **Built** a full-stack price and stock tracker that scrapes a dynamically rendered storefront with **Playwright** and stores history in **Supabase PostgreSQL**, served through an **Express REST API**.
- Automated scraping with a scheduled cycle triggered every 2 hours through an authenticated endpoint; added **retries (up to 3)** and per-attempt logging of retried and failed runs, and verified the flow in production.
- **Developed** a React dashboard for product search, per-option price history, scrape logs, and CSV export; deployed the frontend on **Vercel** and the backend on **Render**.

AutoEIT-II | *Python, PyTorch, SBERT* | GitHub **March 2026**

- **Engineered** a grading pipeline using **SBERT embeddings** and PyTorch to automate semantic evaluation of Elicited Imitation Task responses in Spanish, English, and French across **500+ student samples**.
- Owned the full pipeline preprocessing, embedding generation, inference cutting manual evaluation effort by **˜40%** while improving scoring consistency.

YT Vision Explainer | *JS, Chrome API, Gemini Vision API, Shadow DOM* | GitHub | Demo **May 2026**

- **Designed** a Chrome extension using Gemini Vision and Chrome APIs that captures YouTube frames and explains on-screen code tutorials in real time, no tab-switching needed.
- Used **Shadow DOM** to isolate the extension UI without breaking YouTube's native layout; got frame-to-explanation latency down to **1.5–2s**.

Achievements

Top 5 Finalist – Smart Bin IoT System | *Arduino, Flask, IoT* | GitHub | Demo **Aug 2025**

- **Prototyped** a real-time data-collection system with ESP-32 sensors and a **Flask API** that alerts sanitation teams when a bin is full. Placed **Top 5 of 300 teams** at a national hackathon.

Education

Bachelor of Information Technology **Aug 2023 – May 2027**
Bharati Vidyapeeth's College of Engineering, New Delhi *CGPA: 8.9*

Class XII (CBSE) **Mar 2022**
Rotary Public School *87%*

Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL
Backend & Frameworks: FastAPI, Django, Flask, Node.js, Express, React, REST APIs
ML/Data: PyTorch, scikit-learn, Pandas, NumPy, SBERT, NLP, Computer Vision
Tools: Git, GitHub, GitHub Actions (CI), Docker, PostgreSQL, SQLite, Playwright, Chrome Extension API
Competitive Programming: LeetCode | GeeksforGeeks (Data Structures & Algorithms)